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Date: _____

Quantum Physics S/Q

Solved S/Q Fall-2022 Day: _____

Q.1 Express Plank radiation formula in terms of wave length?

Ans.

$$R_{(\lambda)} = \frac{2\pi c^2 h}{\lambda^5} \left[e^{\frac{hc}{\lambda kT}} - 1 \right]^{-1}$$

where h is Plank's Constant

c is velocity of light

λ is wave length of thermal radiation

k is Boltzmann's Constant

and T is absolute Temperature.

Q.2 Find the energy of a 700nm photon?

Ans.

As we know

$$E = \frac{hc}{\lambda}$$

$$\lambda = 700 \text{ nm} = 700 \times 10^{-9} \text{ m}$$

$$c = 3 \times 10^8 \text{ m s}^{-1}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

Putting values, $E = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{700 \times 10^{-9}}$

$$E = 2.83 \times 10^{-19} \text{ J}$$

Q.3 What voltage must be applied to an X-ray tube for it to emit X-rays with a minimum wavelength of 30pm?

Ans.

Formula:

$$E = \frac{hc}{\lambda}$$

$$\lambda = 30 \text{ pm} = 30 \times 10^{-12} \text{ m}$$

$$eV = \frac{hc}{\lambda}$$

$$V = \frac{hc}{e\lambda}$$

$$V = 4.13 \times 10^4 \text{ Volts}$$

Date: _____

(2)

Day: _____

Q.4) The distance b/w adjacent atomic planes in Calcite (CaCO_3) is 0.300 nm . Find the smallest angle of Bragg scattering for 0.030 nm X-rays.

Ans. As Bragg's equation is

$$2d \sin \theta = n\lambda$$

For smallest angle, $n=1$

Here $d = 0.300 \text{ nm} = 0.300 \times 10^{-9} \text{ m}$

$\lambda = 0.030 \text{ nm} = 0.030 \times 10^{-9} \text{ m}$

Putting Values :- $2 \times 0.300 \times 10^{-9} \times \sin \theta = (1) \times 0.030 \times 10^{-9}$

$\sin \theta = 0.05$

$\theta = \sin^{-1}(0.05)$

$\theta = 2.9^\circ$

Q.5) What is the frequency of an X-ray photon whose momentum is $1.1 \times 10^{-23} \text{ kg m s}^{-1}$?

Ans From De-Broglie Hypothesis

$$\lambda = \frac{h}{p}$$

Here $p = 1.1 \times 10^{-23} \text{ kg m s}^{-1}$

$$\lambda = \frac{6.63 \times 10^{-34}}{1.1 \times 10^{-23}}$$

$\lambda = 6.06 \times 10^{-11} \text{ m}$

Now $f = \nu = ?$

We know

$c = \nu \lambda$

$\{ \nu = \frac{c}{\lambda}$

$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{6.06 \times 10^{-11}}$

$c = \nu \lambda$

$\nu = 4.9 \times 10^{18} \text{ Hz}$

(3)

Date: _____

Day: _____

Q.6 A positron collides head on with an electron and both are annihilated. Each particle had a K.E. of 1.0 MeV. Find the wavelength of resulting photons.

Ans. For Annihilation Process $e^+ + e^- \rightarrow \gamma + \gamma$

Total Energy of Positron = 1 MeV + 0.51 MeV = 1.51 MeV

" " " electron = 1 MeV + 0.51 MeV = 1.51 MeV

Total Energy = 1.51 + 1.51 = 3.02 MeV

As Energy is distributed among two γ ray photons,

So Energy of each photon = $\frac{3.02}{2} = 1.51 \text{ MeV}$

$E = 1.51 \text{ MeV}$

(1 eV = $1.6 \times 10^{-19} \text{ J}$) $E = 1.51 \times 10^6 \text{ eV} = 1.51 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$

As $E = \frac{hc}{\lambda}$

$$\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.51 \times 10^6 \times 1.6 \times 10^{-19}}$$

$$\lambda = 8.21 \times 10^{-13} \text{ m.}$$

Q.7 Find the de-Broglie wavelength of a 1.0 mg grain of sand blown by the wind at a speed of 20 m s^{-1} ?

Ans. From de-Broglie Hypothesis,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Here $m = 1.0 \text{ mg} = 1 \times 10^{-3} \text{ kg}$

$v = 20 \text{ m s}^{-1}$

(4)

Date: _____

Day: _____

$$\lambda = \frac{6.63 \times 10^{-34}}{1 \times 10^{-3} \times 20}$$

$$\lambda = \text{----- m}$$

Q. 8 What effect on the scattering angle in the Davisson-Germer experiment does increasing the electron energy

Ans. As we know $2d \sin \theta = n\lambda$

$$\sin \theta = \frac{n\lambda}{2d} \quad \text{--- (1)}$$

From de-Broglie Hypothesis

$$\lambda = \frac{h}{p} \quad \text{--- (2)}$$

$$\text{As } E = \frac{1}{2}mv^2$$

$$mE = \frac{1}{2}m^2v^2$$

$$2mE = m^2v^2 \quad (p = mv)$$

$$2mE = p^2$$

$$\Rightarrow p = \sqrt{2mE} \quad \text{--- (3)}$$

$$\text{Put in eq. (2)} \quad \lambda = \frac{h}{\sqrt{2mE}}$$

Now put this value of λ in eq. (1)

$$\sin \theta = \frac{n}{2d} \left(\frac{h}{\sqrt{2mE}} \right)$$

$$\sin \theta = \frac{nh}{2d\sqrt{2mE}}$$

$$\sin \theta \propto \frac{1}{\sqrt{E}} \quad \text{--- (4)}$$

On increasing energy, scattering angle decreases

(5)

Date: _____

Day: _____

Q.9 A proton in a one dimensional box has an energy of 400 keV in its first excited state. What is the width of the box?

Ans. Energy - 400 keV = $400 \times 10^3 \text{ eV}$
 $= 400 \times 10^3 \times 1.6 \times 10^{-19} \text{ J}$

Mass of proton = $m = 1.67 \times 10^{-27} \text{ kg}$

For 1st excited state, $n = 2$

As we know $E_n = \frac{n^2 h^2}{8mL^2}$

$\Rightarrow L^2 = \frac{n^2 h^2}{8mE_n}$

$L = \sqrt{\frac{(2)^2 \times (6.63 \times 10^{-34} \text{ Js})^2}{8 \times 1.67 \times 10^{-27} \times 400 \times 10^3 \times 1.6 \times 10^{-19} \text{ C}}}$

$L = 4.52 \times 10^{-14} \text{ m}$

Q.10) What is the shortest wavelength present in the Brackett-series of spectral lines?

Ans. From Brackett Series,

$\frac{1}{\lambda} = R_H \left[\frac{1}{4^2} - \frac{1}{n^2} \right]$

For shortest Wavelength, $n = \infty$

$\frac{1}{\lambda} = R_H \left[\frac{1}{16} - \frac{1}{\infty} \right]$

$\frac{1}{\lambda} = R_H \left(\frac{1}{16} - 0 \right)$

$\left(\frac{1}{\infty} = 0 \right)$

Date: _____

(6)

Day: _____

$$\frac{1}{\lambda} = \frac{R_H}{16}$$

$$\Rightarrow \lambda = \frac{16}{R_H}$$

$$[R_H = 1.097 \times 10^7 \text{ m}^{-1}]$$

$$\lambda = \frac{16}{1.097 \times 10^7}$$

$$\lambda = 1.46 \times 10^{-6} \text{ m}$$

$$\lambda = 1.46 \mu\text{m}$$

Q.11) In the Bohr's model, the electron is in constant motion. How can such an electron have a negative amount of energy?

Ans:- If the electron is in constant motion, it has K.E. The K.E should be positive here. Hence potential energy could be negative.

The total energy i.e. sum of positive K.E and negative potential energy must be negative. Because, for a particle subjected to an inverse square attractive force, the P.E is twice the K.E.

Hence, the electron has a negative amount of energy.

Q.12) Under what circumstances, if any, L_z is equal to L ?

Ans. The total angular momentum of an electron is

$$L = \sqrt{l(l+1)}\hbar$$

$$\text{Here } l = 0, 1, 2, \dots, (n-1)$$

(7)

Date: _____

Day: _____

And $L_z = m_z \hbar$

Here $m_z = 0, \pm 1, \pm 2, \pm 3, \dots, \pm l$

This means L_z is integral multiple of $\hbar$

But we have $L \leq \sqrt{l(l+1)} \hbar$

$$\Rightarrow L^2 \leq l(l+1) \hbar^2$$

Equality sign happens when $l = 0$

$\therefore L$ and L_z are equal for $l = 0$.

$$L = L_z = 0 \text{ when } l = 0$$

Q.13) A beam of 13.0 eV electrons is used to bombard gaseous hydrogen. What series of wavelength will be emitted?

Ans. As the electron beam is having energy of 13 eV, it can excite the atom to the states whose energy is

$$13.6 \text{ eV} - 13 = 0.6 \text{ eV}$$

As $E_4 = 0.85 \text{ eV}$

$$E_5 = 0.544 \text{ eV}$$

So the electron beam can excite the hydrogen gas maximum to 4th energy state. Hence the transit electron can come back to ground state from either of three excited state, thus emitting Lyman, Balmer and Paschen Series.

Q.14 The azimuthal wave function of hydrogen atom is

$$F(\phi) = A e^{im\phi}$$

Integrate $|F|^2$ over all values from 0 to 2π and obtain the value of normalization constant A .

Ans. The azimuthal wave function for hydrogen atom is

$$F(\phi) = A e^{im\phi} \quad \text{--- (1)}$$

$$\text{and } F^*(\phi) = A e^{-im\phi} \quad \text{--- (2)}$$

On Normalization of wave function

$$\int_0^{2\pi} F^*(\phi) F(\phi) d\phi = 1$$

From eq. (1) and (2)

$$\int_0^{2\pi} (A e^{-im\phi}) (A e^{im\phi}) d\phi = 1$$

$$\int_0^{2\pi} A^2 e^{-im\phi + im\phi} d\phi = 1$$

$$A^2 \int_0^{2\pi} d\phi = 1$$

$$A^2 \left| \phi \right|_0^{2\pi} = 1$$

$$A^2 (2\pi) = 1$$

$$A^2 = \frac{1}{2\pi}$$

$$A = \sqrt{\frac{1}{2\pi}}$$

Q.15) Why is the ground state of the Hydrogen atom not split into two sublevels by spin-orbit coupling?

Date: _____

(9)

Day: _____

Ans. Ground state electronic configuration of H-atom is $1s^1$.

So, it has only one electron in s state, in which there is no orbital angular momentum i.e.

$$l=0 \text{ and } s=\frac{1}{2}$$

Depending upon the orientation of its spin vector

' s ', for single electron $s=\frac{1}{2}$. So only one sub state will exist i.e. it can't split into two sublevels.

THE END.

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